2.5 Common file formats in QGIS

A **format** is the way in which geodata are stored in a file or database. You can usually recognise the format of a file by the **file extension**: the letters after the last dot in the file name (`.gpkg`, `.shp`, `.tif`…).

There are many formats for geodata. This page explains four of the formats you will meet most often in QGIS and in this learning path.

Format	Extension	Data	Typical use
GeoPackage	`.gpkg`	vector (and raster)	the default format of QGIS for vector data; one file can hold several layers
Shapefile	`.shp` (+ other files)	vector	an older but very common format for exchanging vector data; almost every GIS program can read it
GeoTIFF	`.tif`, `.tiff`	raster	scanned and georeferenced maps, aerial photographs, elevation models…
JPEG 2000	`.jp2`	raster	like GeoTIFF, but compressed (smaller files); often used for large aerial photographs

GeoPackage: the default for vector data

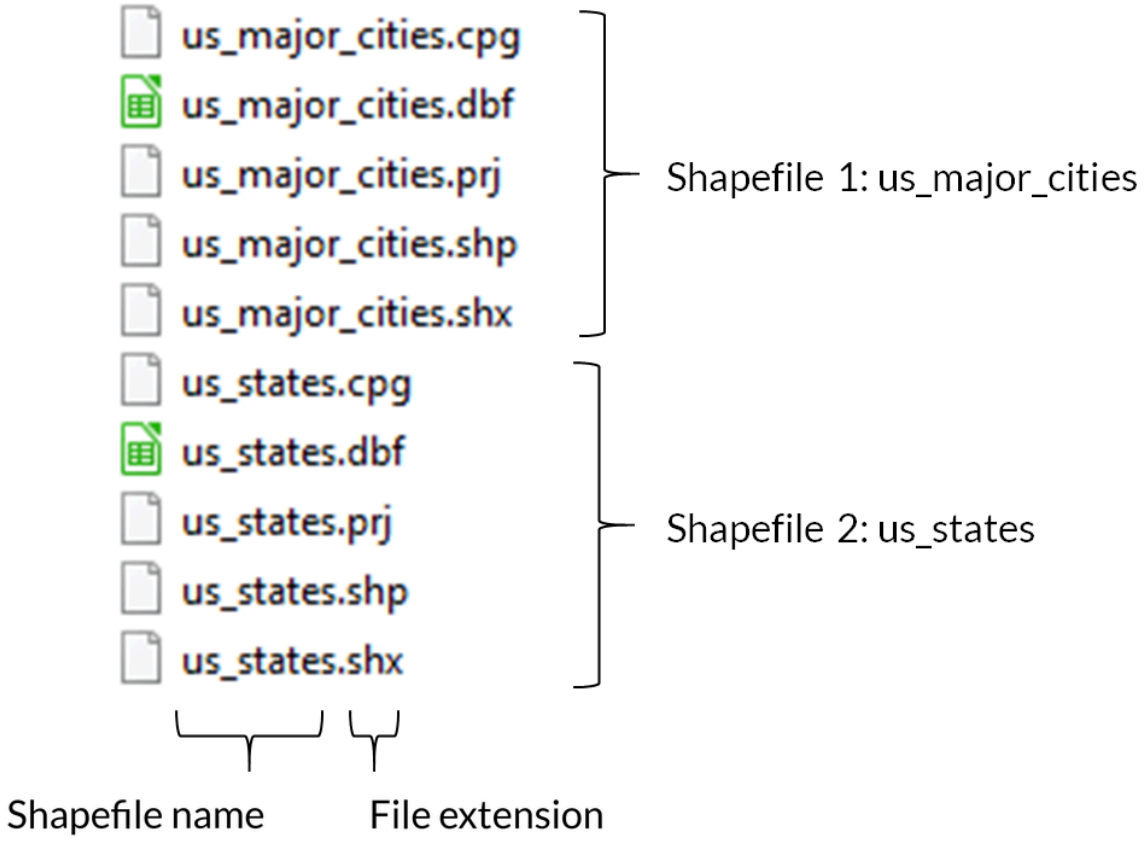
QGIS uses the **GeoPackage** format (`.gpkg`) as its default format for vector data. A GeoPackage is a **single file** that can hold **several layers** (for example the points, lines and polygons of one project), and even rasters and plain tables. You can store not only the data themselves (for example the locations of points), but also how they are shown in GIS (for example as red stars). Because everything is in one file, a GeoPackage is easy to copy or send.

How you save several layers in one GeoPackage, convert shapefiles and save how your layers are shown is explained later, in Working with GeoPackages.

Shapefile: older, but still everywhere

The **shapefile** (or Esri shapefile) dates from the 1990s, but it is still one of the most widely used vector formats: almost every GIS program can read it, and a lot of existing data are offered in it. In this learning path you will often *open* shapefiles, but when you *save* or *export* your own data, we use GeoPackage.

A shapefile is a **combination of several files** (at least three).



A shapefile always consists of:

- **SHP file**: spatial data component
- **DBF file**: attribute information
- **SHX file**: an index of the geometries in the SHP file

A shapefile can also include several extra files, such as:

- **PRJ file**: information about the coordinate reference system used
- **SHP.XML file**: metadata
- **CPG file**: information about character encoding

> Warning
>
> **The files that form a shapefile must always be stored in the same folder with the same file name** (only the extension differs, e.g. `us_states.shp`, `us_states.dbf`, `us_states.shx`). When you copy, move or rename a shapefile, always do it with all its files.

Compared with a GeoPackage, a shapefile has some limits:

- One shapefile holds only **one layer** with one geometry type (only points, only lines or only polygons).
- It stores only the data, **not how they are shown** in GIS.

GeoPackage vs shapefile

	GeoPackage	Shapefile
Files	one `.gpkg` file	several files that must stay together

	GeoPackage	Shapefile
Layers	several per file	one per shapefile
How the data are shown	can be stored in the file	not stored
Raster data	yes	no
Compatibility	most modern GIS software	almost all GIS software

Raster files: GeoTIFF and JPEG 2000

GeoTIFF (`.tif` or `.tiff`) is the most common format for raster data. Besides the pixel values, a GeoTIFF stores **where on earth the image belongs**: the coordinates of the image and its coordinate system. That is why QGIS places a GeoTIFF in the right position at once.

> Warning
>
> **Not every `.tif` file is a georeferenced raster.** A scan of an old map saved as TIFF is just an image, without coordinates. You first have to give it a position on the earth yourself; this is explained in the chapter Georeferencing.

JPEG 2000 (`.jp2`) files have similar properties to GeoTIFFs for storing raster data, but they are compressed, so the files are much smaller. That is why governments and mapping agencies often use them for large aerial photographs (orthophotos). On the other hand, not every GIS program can open JPEG 2000 files, and drawing them on screen is often slower. When in doubt, GeoTIFF is the safer choice.

Other formats

This page discussed the formats you will meet most often, but there are many more. Some of them, such as CSV, GeoJSON, KML and the Esri file geodatabase, are explained in the chapter Data in GIS (part 2).